

# ARSHAM VASHISTH

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[LinkedIn](#) 

[Portfolio](#) 

DATA SCIENCE | MACHINE LEARNING | AI

## MOTIVATION

*Data Scientist with a Master of Artificial Intelligence degree from RMIT University and hands-on experience developing machine learning solutions using Python. Experienced in data preparation, predictive modelling, statistical analysis, and model evaluation through research, academic projects, and a Machine Learning Engineering internship at Silverpond. Passionate about applying AI to solve real-world problems and improving outcomes through data-driven research.*

## SKILLS & TOOLS

**Programming:** SQL, Python, R, Java, C, C++

**Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, CNNs, Computer Vision, Feature Engineering, Model Evaluation, Hyperparameter Tuning

**Data Science & Statistics:** Predictive Modelling, Classification, Regression, Clustering, PCA, Association Rule Learning, A/B Testing, Hypothesis Testing, Causal Impact Analysis, Statistical Analysis

**Tools & Technologies:** Git, Jupyter Notebook, Anaconda, Scikit-learn

**Mathematics:** Linear Algebra, Probability, Statistics, Experimental Design

## PROJECTS

### **Enhancing Targeting Accuracy Using Machine Learning**

- By analyzing historical customer behavior and campaign data, developed a [machine learning](#) model to predict the likelihood of customers signing up for a \$100 annual membership. This enabled the retailer to identify high conversion customers, reduce mailer distribution costs, and significantly [improve return on investment \(ROI\)](#). The solution provided insights for more targeted, cost-effective marketing strategies.

### **Predicting Customer Loyalty Using Machine Learning**

- Led the development of a predictive model to estimate customer loyalty scores for a grocery retailer's untapped customer base. After market-level loyalty [data was only available for 50% of the customer database](#), I built a machine learning solution to identify key [relationships between customer behavior and loyalty metrics](#). This enabled the retailer to predict accurate loyalty scores for the remaining customers, ensuring [better customer segmentation, targeted marketing, and more effective communication](#) strategies ultimately improving customer retention and engagement.

### **Assessing Campaign Performance Using Chi-Square Test For Independence**

- Conducted an [A/B test](#) for a grocery retailer's promotional campaign of their new "Delivery Club" membership program. Customers were randomly divided into three groups: low-cost mailer, high-cost mailer, and a control group. [Analyzed the differences in signup rates between the mailer types](#) to determine the most effective marketing strategy. The insights generated provided the client with clear recommendations for optimizing future campaigns, allowing for smarter marketing spend allocation and improved ROI.

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## PROJECTS (CON'T)

### **Global Earthquake Tracking Dashboard (Tableau)**

- Developed an interactive Tableau dashboard that visualizes [global earthquake activity over a 30 day period](#). The dashboard enables users to explore key patterns, magnitudes, and regional distributions of seismic events. By focusing on intuitive design and visually engaging features, it provides quick insights into earthquake [frequency, intensity, and geographical trends](#), empowering users to analyze and interpret seismic data with ease.

## EXPERIENCE

### **Machine Learning Engineer, Intern - Silverpond**

JUNE 2025 - DEC 2025

- To [enhance perception capabilities](#), I contributed to Highlighter by developing and [optimizing computer vision models](#), improving overall model performance by ~18% across multiple use cases.
- To improve [object detection and segmentation](#) performance, I trained YOLO-based models in Python, [increasing detection accuracy by 22%](#) and reducing inference time by 15% in production environments.
- To enable precise agricultural analytics, I built a semantic segmentation model for chicken beak detection, [achieving an IoU score of 0.87 and improving annotation efficiency by 30%](#).
- To strengthen urban scene understanding, I developed and fine-tuned models for trains and trams, [improving object recognition accuracy by 20%](#) in complex real-world environments.
- To support scalable deployment, I collaborated with cross-functional teams to integrate machine learning solutions, [reducing model deployment time by 25% across diverse datasets and pipelines](#).

## EDUCATION

### **Master of Artificial Intelligence**

2021 - 2023 - RMIT University

### **Bachelors of Computer Science and Engineering**

2016 - 2020 - SRM University

## REFERENCES

### **Jonathan Chang**

Chief Technology Officer

Silverpond

[jonathan.chang@silverpond.com.au](mailto:jonathan.chang@silverpond.com.au)

### **Joshua Patterson**

Machine Learning Engineer

Silverpond

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